

Measuring Housing Affordability

October 1, 2026

Congressional Research Service

<https://crsreports.congress.gov>

R49457



Measuring Housing Affordability

Housing affordability is an area of interest for policymakers and the public. There are a variety of ways to measure housing affordability, with the most common being the housing-cost-to-income ratio approach (hereinafter, “ratio approach”). This approach considers housing “unaffordable” if housing costs exceed 30% of household income. Recently, the residual income approach—which defines housing as “affordable” if the difference between household income and housing costs is sufficient to pay for some socially acceptable basket of non-housing necessities—has gained attention from researchers due to its ability to address some of the shortcomings of the ratio approach. Understanding the conceptual and empirical differences between these approaches is important, because the concept of what is affordable influences subsidy levels in federal assistance programs, research on family economic well-being, and reports on the health of the nation’s housing market.

This report compares the ratio and residual income approaches to measuring housing affordability. While these two measures often show similar overarching housing affordability patterns, they can generate different estimates of the *extent* of affordability challenges for certain subpopulations. Using data from the 2023 American Community Survey, CRS found the following among non-elderly U.S. households:

- Compared to the residual income approach, the ratio approach estimates *lower* rates of housing affordability challenges for all households, low-income households, and households with children.
- Approximately 53.0% of households did not face housing affordability challenges under either approach, 31.8% faced challenges under both approaches, and 15.2% faced challenges according to only one of the approaches.
- Under the residual income approach, it would require more than three times as much additional household income—from work or public subsidies—to alleviate housing affordability challenges than is required under the ratio approach.

CRS analysis of the 2023 American Housing Survey found that the two approaches produce similar results when estimating the percentage of households facing various indicators of housing insecurity (e.g., difficulty affording housing payments, threat of eviction). Additionally, both approaches found that households with housing affordability challenges were approximately twice as likely as those without such challenges to report having difficulty making housing payments and experiencing involuntary moves. Thus, even though the housing-cost-to-income ratio may not fully capture the prevalence of housing affordability challenges in the United States, it does serve to reflect households’ experiences with housing insecurity.

R49457

October 1, 2026

Mary Daniels
Innovation Policy Lab
Fellow

Mark P. Keightley
Specialist in Economics

Maggie McCarty
Specialist in Housing Policy

Contents

Introduction	1
The Ratio Approach.....	2
Advantages and Disadvantages	3
The Residual Income Approach	4
Advantages and Disadvantages	5
Comparing the Ratio and Residual Income Approaches	6
Cost Burden and Residual Income	7
The Relationship Between Cost Burden and Shelter Poverty	8
The Prevalence of Housing Affordability Challenges.....	9
Housing Affordability for Different Populations	10
The Size of the Affordability Gap	14
Housing Cost Levels “Affordable” to Households	15
Housing Insecurity	17
Households’ Overall Economic Well-Being	19
Policy Considerations.....	21

Figures

Figure 1. How Shelter Poverty Rates Vary by Cost Burden.....	8
Figure 2. Percentage of Non-Elderly Households with Housing Affordability Challenges, by Household Characteristics	11
Figure 3. Percentage of Non-Elderly Households with Housing Affordability Challenges, by Household Income.....	12
Figure C-1. Percentage of Non-Elderly Households with Housing Affordability Challenges, by State	32
Figure C-2. Characteristics Correlated with the Gap Between a State’s Housing Cost Burden and Shelter Poverty Rates.....	33

Tables

Table 1. Cost Burden and Residual Income for Non-Elderly Households	7
Table 2. Non-Elderly Households Experiencing Housing Cost Burden, Shelter Poverty, Neither, or Both in 2023	10
Table 3. Distribution of Non-Elderly Households with Affordability Challenges	14
Table 4. Affordability Gap for Non-Elderly Households: 2023	15
Table 5. Annual Housing Costs Considered “Affordable,” by Affordability Approach and Income	16
Table 6. Percentage of Non-Elderly Households Experiencing Indicators of Housing Insecurity	18
Table 7. 2023	18

Table B-1. The Prevalence of Housing Affordability Challenges Among Non-Elderly U.S.
Households After Accounting for Subsidies and Public Transportation 29

Table D-1. Housing Affordability Indicators..... 35

Appendixes

Appendix A. Methodology 23

Appendix B. Sensitivity Analyses 27

Appendix C. Geographic Differences in Affordability Estimates 31

Appendix D. Housing Affordability Measures..... 34

Contacts

Author Information..... 38

Introduction

Housing affordability is an area of interest for policymakers and the public. There are a variety of methods for measuring housing affordability, which can vary by their mathematical approach (a ratio versus a difference), whether they establish benchmarks for housing and non-housing consumption, and how they define *housing costs* and *household income*.¹ While numerous housing affordability measures exist, housing is generally considered affordable if its costs do not exceed 30% of household income. The 30% affordability standard is the most commonly used housing affordability measure and has been incorporated in the design of the nation's largest housing assistance programs, such as the Housing Choice Voucher, public housing, Section 8 project-based rental assistance, and Low-Income Housing Tax Credit programs.²

In addition to influencing the design of federal rental assistance programs, the concept of housing affordability has been used to analyze individual well-being and housing market conditions. It is used as a predictor of household well-being, with research showing that households that devote a relatively large share of income toward housing costs are less likely to invest in enrichment activities for their children and more likely to experience material hardship.³ Trends in housing affordability are also used to describe the health of the nation's housing markets in reports such as the U.S. Department of Housing and Urban Development's (HUD's) *Worst Case Housing Needs* series of reports to Congress and the Joint Center for Housing Studies' annual *State of the Nation's Housing* reports.

Given the importance of housing affordability in federal policy, this CRS report describes how housing affordability is conceptualized and provides analysis of two measures of affordability: the commonly used housing-cost-to-income ratio approach (hereinafter, "ratio approach") and a residual income approach, which has recently gained more attention from researchers. The report discusses the origins, advantages, and disadvantages of each measure. It then uses the American Community Survey (ACS) and the American Housing Survey (AHS) to analyze how the definition of *housing affordability* affects the understanding of non-elderly households' experiences with housing affordability challenges. Specifically:

- ACS data are used to examine the prevalence and characteristics of non-elderly households with affordability challenges according to each affordability measure. This provided insight into when and for which populations these two measures diverge when estimating housing affordability challenges.
- ACS data are then used to examine how much income is required for housing to be considered affordable and what is considered an affordable housing cost. This can provide context for considering potential housing subsidy levels.
- AHS data are used to examine the overlap between the households that self-report housing insecurity and those estimated to be facing housing affordability challenges by the ratio and residual income approaches. This provides information on how well the numerical measures of affordability can identify households that are struggling with housing insecurity.

¹ See **Table D-1** for an illustrative list of housing affordability measures.

² For more information, see CRS Report RL34591, *Overview of Federal Housing Assistance Programs and Policy*, by Maggie McCarty, Libby Perl, and Katie Jones; and CRS Report RS22389, *An Introduction to the Low-Income Housing Tax Credit*, by Mark P. Keightley.

³ Sandra J. Newman and C. Scott Holupka, "Housing Affordability and Investments in Children," *Journal of Housing Economics*, vol. 24 (June 2014), pp. 89-100; and Shomon Shamsuddin and Colin Campbell, "Housing Cost Burden, Material Hardship, and Well-Being," *Housing Policy Debate*, vol. 32, no. 3 (2022), pp. 413-432.

- Lastly, ACS data are used to determine which basic necessities in a recommended family budget non-elderly households could afford to purchase in 2023 given their reported incomes and housing costs. This can help inform how high housing costs affect overall economic well-being.

The report concludes with a discussion of policy considerations.

The Ratio Approach

The ratio approach has its origins in 19th century statistical research on household expenditure patterns. Researchers observed that households typically spent 20%-25% of household income on housing costs.⁴ This became the general rule of thumb for how much households should spend on housing costs and would be adopted by mortgage lenders, private landlords, academics, and federal agencies.⁵ Over time, the ratio approach would become a staple in federal policymaking, influencing eligibility requirements and subsidy levels in rental assistance programs, the underwriting of mortgage loans, and national estimates of housing affordability. The standard would also evolve from spending 20%-25% of income on housing to the 30% rule used today. The timeline below describes selected moments in this development.

- **1937.** The United States Housing Act of 1937 (P.L. 75-412) limited eligibility for public housing to households whose incomes did not exceed five times the rent charged.
- **1938.** An amendment to the National Housing Act of 1934 (P.L. 90-448) used a ratio approach to determine the subsidy level for mortgage assistance provided to low-income homeowners.
- **1969.** The so-called Brooke Amendment to the Housing and Urban Development Act of 1969 (P.L. 91-152) capped the amount of rent families in public housing would be required to pay at 25% of adjusted family income.
- **1979.** The Housing and Community Development Amendments of 1979 (P.L. 96-153) increased rents in public housing and the Section 8 programs to 30% of adjusted family income for all families except very-low-income families.
- **1981.** The Housing and Community Development Amendments of 1981 (P.L. 97-35) increased rents for *all families* in public housing and the Section 8 programs to 30% of adjusted family income.
- **1990.** The Cranston-Gonzalez National Affordable Housing Act (P.L. 101-625) required states and local jurisdictions to submit a comprehensive housing affordability strategy (CHAS) to receive funding from certain federal housing programs. To assist state and local governments in complying with this requirement, HUD created the CHAS database, which uses the 30% affordability standard to characterize housing affordability patterns.
- **1991.** HUD published the first *Worst Case Housing Needs* report to Congress.⁶ This report (and subsequent reports) described families as having worst case

⁴ J. David Hulchanski, "The Concept of Housing Affordability: Six Contemporary Uses of the Housing Expenditure-to-Income Ratio," *Housing Studies*, vol. 10, no. 4 (1995), pp. 471-491. When calculating housing costs, researchers generally include rent/mortgage, utilities, and other homeowner costs such as real estate taxes and insurance.

⁵ Hulchanski, "The Concept of Housing Affordability," p. 475; and Danilo Pelletiere, "Getting to the Heart of Housing's Fundamental Question: How Much Can a Family Afford? A Primer on Housing Affordability Standards in U.S. Housing Policy," National Low Income Housing Coalition, February 2008, pp. 1-6, <https://nlihc.org/resource/getting-heart-housings-fundamental-question-how-much-can-family-afford>.

⁶ HUD, *Priority Housing Problems and "Worst Case" Needs in 1989*, June 1991, p. 1, https://www.huduser.gov/portal/publications/affhsg/wc_HsgNeeds89_report.html.

housing needs if they lived in substandard housing conditions and/or devoted more than 50% of their income toward housing.

Advantages and Disadvantages

Using a ratio approach to measure housing affordability has several advantages over the residual income approach. Private organizations, academic researchers, and government agencies have consistently used it over time, making it familiar and generally well understood. The ratio is also simple to calculate and uses information (housing costs and household income) available in reliable public databases (e.g., Census Bureau surveys), resulting in consistent estimates of housing affordability among researchers. The use of a ratio also makes it possible to compare housing affordability patterns across time and geographies without needing to make adjustments (e.g., adjusting for inflation or price parities).

There are questions about which threshold to use to define housing affordability. This decision has historically been based on observed household expenditure patterns rather than theory about what ratio increases the likelihood of adverse outcomes. While researchers have tried to pinpoint the threshold at which adverse outcomes become more likely, there is no consensus. Some research has found that for households with incomes below 200% of the federal poverty line, spending on child enrichment and children's cognitive performance are maximized when spending 30%-35% of income on housing.⁷ Other scholars found that the likelihood of experiencing material hardship (e.g., food insecurity, difficulty paying bills, delaying medical care) continued increasing until households were spending around 50% of income on housing.⁸

The ratio approach does not account for household characteristics that influence affordability, such as household income and composition. For example, a household earning \$150,000 annually that spends 45% of income on housing is considered cost-burdened, while a household that earns \$20,000 a year and spends 25% of income on housing is not considered cost-burdened. However, the higher-income household is more likely to be able to cover its basic needs with its post-housing income than the lower-income household is. The ratio approach also implies that a single-person household earning \$40,000 requires the same amount of money to pay for non-housing necessities as an adult with two children earning the same amount. However, the parent likely has additional costs, such as child care, that require more expenditures on non-housing goods than the single person does.

Defining *affordability* with a ratio can also make it difficult to distinguish between preferences and a lack of alternative housing options. When deciding where to live, people in households consider more than the cost of housing. They also consider the bundles of features and services that housing provides access to, such as the number of bedrooms and bathrooms, neighborhood amenities, the quality of the school district, access to public transportation, commuting distance, proximity to family, and more. Preferences about these characteristics may lead a household to choose a housing unit that costs more than 30% of its income. Thus, it is not always clear if a household spending more than 30% of its income on housing is doing so because a cheaper unit is not available or because the unit provides additional benefits that the household values.

⁷ Sandra J. Newman and C. Scott Holupka, "Housing Affordability and Investments in Children," *Journal of Housing Economics*, vol. 24 (June 2014), pp. 89-100; and Sandra J. Newman and C. Scott Holupka, "Housing Affordability and Child Well-Being," *Housing Policy Debate*, vol. 25, no. 1 (2015), pp. 116-151.

⁸ Shomon Shamsuddin and Colin Campbell, "Housing Cost Burden, Material Hardship, and Well-Being," *Housing Policy Debate*, vol. 32, no. 3 (2022), pp. 413-432. The paper found that once housing costs as a percentage of income exceeded 50%, there was little additional increase in material hardship as housing costs continued to increase, indicating that severely cost-burdened households are so financially strained that additional spending on housing has marginal effects on their financial situations.

The Residual Income Approach

Recognizing the shortcomings of the ratio approach, U.S. housing analysts in the 1960s and early 1970s sought to find an alternative method for describing the financial strain induced by high housing costs. From this scholarship arose the residual income approach to measuring housing affordability. In a 1966 pamphlet, housing activist Cushing Dolbeare proposed that the residual income approach be used to set subsidy levels for housing assistance.⁹ In 1971, housing experts submitted reports to the U.S. House Committee on Banking and Currency recommending that residual income be considered when setting federal housing subsidies and analyzing housing need.¹⁰ Academic researchers later analyzed housing affordability patterns in the United States using this new metric.¹¹

Rather than relating housing and non-housing expenditures to household resources via a ratio, the residual income approach examines the difference between resources and expenditures. This metric calculates the difference between a household's income and housing costs to determine its "residual income." Affordability is then assessed based on whether residual income is greater than or equal to the amount needed to purchase some combination of non-housing goods, with households that fail to meet this standard being described as "shelter poor."¹²

While not nearly as widely used as the ratio approach, the residual income approach has been included in federal regulations. When underwriting loans to veterans under the U.S. Department of Veterans Affairs' Loan Guaranty Program, lenders must generally assess both the borrower's debt-to-income ratio and residual income to determine if the veteran has enough present and anticipated income to qualify for the loan.¹³ These regulations implement Section 402 of the Veterans' Benefits Improvement and Health Care Authorization Act of 1986 (P.L. 99-576), and the residual income guidelines were last updated in 1997.¹⁴ Other federal mortgage programs also require lenders to evaluate borrowers' residual income as part of the program's underwriting process. For example, since September 19, 2017, HUD regulations for the Home Equity Conversion Mortgage program have generally required mortgage lenders to conduct residual income analysis as part of their financial assessments of potential borrowers.¹⁵ Since January 10, 2014, the Consumer Financial Protection Bureau's regulations implementing the Truth in Lending Act of 1968 have also required creditors to consider consumers' monthly debt-to-income ratio or residual income when determining their repayment ability for certain residential mortgage loans.¹⁶

⁹ See Michael E. Stone, "What Is Housing Affordability? The Case for the Residual Income Approach," *Housing Policy Debate*, vol. 17, no. 1 (2006), p. 165.

¹⁰ Ira S. Lowry, "Housing Assistance for Low-Income Urban Families: A Fresh Approach," *Papers Submitted to the Subcommittee on Housing Panels, U.S. House Committee on Banking and Currency, 92nd Cong., 1st sess.*, 1971, pp. 489-523; and Dorothy K. Newman, "Housing the Poor and the Shelter-to-Income Ratio," *Papers Submitted to the Subcommittee on Housing Panels, U.S. House Committee on Banking and Currency, 92nd Cong., 1st sess.*, 1971, pp. 555-578.

¹¹ Michael E. Stone, "Shelter Poverty: The Chronic Crisis of Housing Affordability," *New England Journal of Public Policy*, vol. 20, no. 1 (2004), pp. 108-119; and Nandinee K. Kutty, "A New Measure of Housing Affordability: Estimates and Analytical Results," *Housing Policy Debate*, vol. 16, no. 1 (2005), pp. 113-142.

¹² Stone, "Shelter Poverty," p. 109.

¹³ 38 C.F.R. §36.4340(d)-(e).

¹⁴ See the October 17, 1997, version of 38 C.F.R. §36.4340(e) at <https://www.govinfo.gov/content/pkg/FR-1997-10-17/pdf/97-27564.pdf> and 38 U.S.C. §3710.

¹⁵ 24 C.F.R. §206.37.

¹⁶ 12 C.F.R. §1026.43(c). See also 15 U.S.C. §1639c(a)(3).

Advantages and Disadvantages

The advantages and disadvantages of the residual income approach vary depending on how it is specified: There are two ways to determine how much households require to purchase non-housing necessities, each with its own benefits and weaknesses.

The first option classifies housing as affordable if a household's residual income exceeds some percentage of the federal poverty threshold. While connecting residual income to the official poverty measure allows the residual income approach to benefit from the familiarity of a widely used measure of economic privation, this method of determining housing affordability results in some of the same long-standing criticisms directed at the official poverty measure.¹⁷ For example, the poverty measure does not account for geographic variation in prices that affects what can be purchased with a household's residual income. Additionally, as with the ratio approach, the choice of what percentage of the poverty threshold to use as the cutoff lacks a strong theoretical basis.

The second option, using a family budget standard to calculate non-housing costs, addresses some of these issues but still has shortcomings. Family budgets determine the amount of money a family needs to purchase a basket of essential goods and services that meet its basic needs (see **Table D-1** for examples). This basket is designed to vary by geography, household size, and household composition. Because family budgets usually vary by geography, this standard is better able to capture local price variations that affect affordability. The family-budget-based residual income measure is described as “a sliding scale of affordability” that varies with household size, composition, income, and geography.¹⁸ This flexibility can generate more accurate estimates of the financial strain faced by certain types of households (i.e., those with many members, those with children, and low-income households) than ratio-based measures can.

This option is limited by its inability to take account of individual preferences. As described earlier, households may have preferences about the features and services linked to particular housing units that led them to spend more on housing than recommended by affordability metrics. Ignoring individual preferences can result in incorrect assessments of affordability problems. This is a shortcoming of both residual income options.

Additionally, the methodology for calculating family budgets varies among researchers and is more complex than the ratio approach and poverty-threshold-based residual income approaches. Researchers must make subjective decisions about which items should be included in the family budget and how to approximate their costs, creating the possibility for inconsistent estimates of family budgets and housing affordability patterns. To demonstrate this, researchers at Columbia University compared four different family budgets:

1. The United Way of Northern New Jersey's Asset Limited Income Constrained Employed (ALICE) Household Survival Budget
2. The Economic Policy Institute's (EPI's) Family Budget Calculator (FBC)
3. The Massachusetts Institute of Technology's (MIT's) Living Wage Calculator
4. The University of Washington's (UW's) Self-Sufficiency Standard

The researchers found that EPI's and MIT's budget estimates were generally higher than the ALICE and UW budget estimates.¹⁹ Specifically, they found that for a two-adult, two-child family in Queens County, NY, EPI's budget (in 2023 dollars) was 28.8% higher than UW's budget (in

¹⁷ For more information, see CRS Report R44780, *An Introduction to Poverty Measurement*, by Joseph Dalaker.

¹⁸ Stone, “What Is Housing Affordability?,” p. 164.

¹⁹ Pamela Joshi et al., “Measuring Economic Needs Beyond Poverty: A Consumer Guide to Family Budget Measure,” Center on Poverty and Social Policy, March 2025, pp. 6-8, <https://povertycenter.columbia.edu/sites/povertycenter.columbia.edu/files/content/Publications/Joint-Report/Family-Budget-Consumer-Guide-CPSP-IECOHD-2025.pdf>.

2023 dollars) and 34.0% higher than the ALICE budget (in 2022 dollars²⁰). For a two-adult, two-child household in Grant County, WA, EPI's budget was 18.6% higher than UW's budget and 21.0% higher than the ALICE budget.

Comparing the Ratio and Residual Income Approaches

While the ratio and residual income approaches have their conceptual shortcomings, they remain common and useful ways to understand housing affordability patterns. They often find similar overarching patterns, but their mathematical differences can generate discrepancies in housing affordability estimates. The remainder of the report seeks to identify when these discrepancies arise.

Housing Affordability Terminology

Cost burdened: Housing costs exceed 30% of household income, used in this report for identifying those with affordability challenges using the ratio approach.

Severely cost burdened: Housing costs exceed 50% of household income, used in this report for identifying those with severe affordability challenges using the ratio approach.

Residual income: The amount of household income left after paying for housing costs.

Shelter poor: Housing and estimated necessary non-housing costs exceed household income, used in this report for identifying those with affordability challenges using a residual income approach. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.

In this report, *cost burdened*, *severely cost burdened*, and *shelter poor* are used as mathematical terms to describe the relationship between household income and housing costs rather than a reflection of how households feel about their ability to afford housing. Discussion of a household's perceived struggles with affording housing are discussed in the "Housing Insecurity" section of the report.

This section uses the 2023 ACS Public Use Microdata Samples (PUMS) to compare estimates of housing affordability generated by the ratio and residual income approaches. CRS obtained estimates of household income, household characteristics, and housing costs from the ACS. For renters, housing costs are gross rents, which includes contract rent and utilities. For homeowners, housing costs include the amount spent on mortgages, real estate taxes, insurance, utilities, fuels, mobile home costs, and condominium fees.²¹ Household income is the total pre-tax income of all household members ages 15 and older in the past 12 months, including earnings, retirement income, and public assistance income.²²

²⁰ ALICE's budget was available only in 2022 dollars at the time the report was published.

²¹ The variable used was "selected monthly owner costs."

²² The Census Bureau asks individuals about eight sources of income: (1) wage or salary income; (2) self-employment income; (3) interest, dividends, net rental income, royalty income, and income from estates and trusts; (4) social security or railroad retirement income, (5) supplemental security income; (6) public assistance income; (7) retirement, survivor, or disability income; and (8) all other income, including unemployment compensation, worker's compensation, Department of Veterans Affairs payments, alimony and child support, periodic income obtained from people outside of the household, military family allotments, and any other periodic income other than earnings. For more information, see the ACS subject definitions at <https://www.census.gov/programs-surveys/acs/technical-documentation/code-lists.html>.

The residual income approach used in this report relies on EPI’s 2024 FBC (which is in 2023 dollars) to calculate non-housing costs.²³ Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.²⁴

To be consistent with the FBC, CRS made several sample restrictions. Because the FBC includes cost estimates only for households with one to two adults and zero to four children, the ACS sample was restricted to include only these households. Households with members age 65 or older were also excluded, because elderly individuals may have special expenditures (e.g., home health care) that are not included in the FBC.²⁵ (More details on the methodology for this analysis can be found in **Appendix A**.)

The following analysis of non-elderly households in the ACS begins with summary statistics on cost burden and residual income. It then discusses the relationship between cost burden and shelter poverty. Next, it presents estimates of the prevalence of housing cost burden and shelter poverty among non-elderly households and differences in affordability challenges across demographic groups. It then discusses the affordability gap—the difference between household income and the income needed for housing to be affordable—for non-elderly households in 2023. This section concludes by examining what constitutes an affordable housing cost under each affordability standard.

Cost Burden and Residual Income

Table 1 reports summary statistics on cost burden (housing costs as a percentage of household income) and residual income (household income left after paying housing costs) for non-elderly households in 2023. Median cost burden was 22.6%, which is below the threshold for being considered cost burdened. Median residual income was \$59,000. This means that after paying for housing costs, half of non-elderly households had less than \$59,000 remaining to pay for non-housing necessities and half had more than \$59,000 to pay for non-housing necessities.

Table 1. Cost Burden and Residual Income for Non-Elderly Households

	25 th Percentile	50 th Percentile (Median)	Mean	75 th Percentile
Cost burden	13.6%	22.6%	31.5%	38.4%
Residual income	\$24,640.0	\$59,000.0	\$87,942.3	\$111,800.0

Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute’s 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Residual income is the amount of household income left after paying for housing costs.

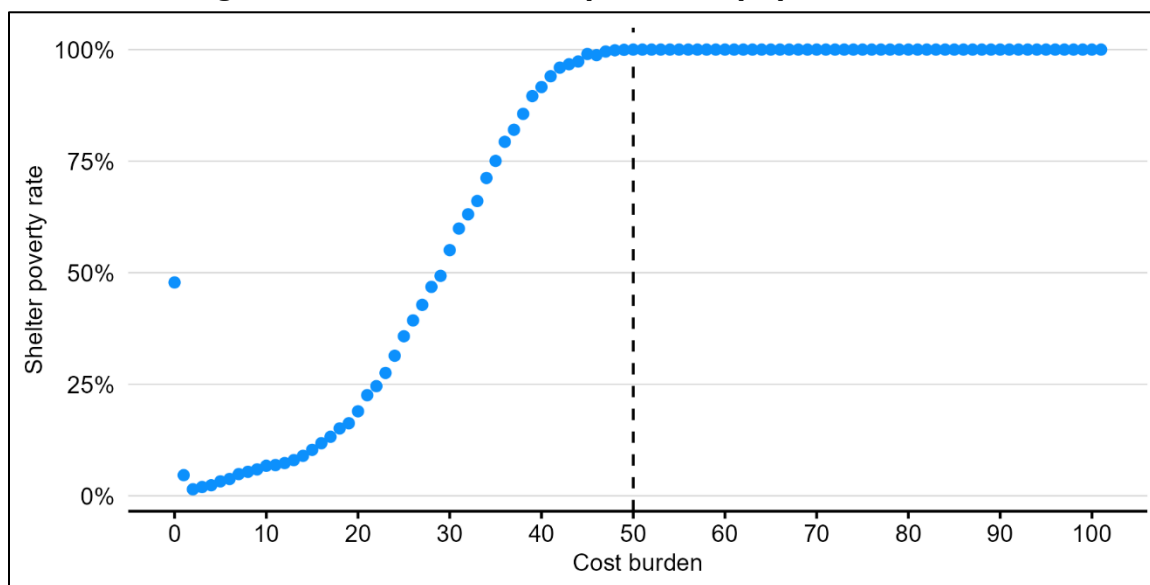
²³ EPI’s family budget was chosen over more conservative family budgets such as United Way’s ALICE and UW’s Self-Sufficiency Standard because EPI has data for all U.S. counties, while the other budgets are missing data for numerous counties. Because EPI’s family budget tends to estimate higher costs than other budgets do, it may result in higher estimates of housing affordability challenges than other family budgets would estimate. In **Appendix B**, CRS conducted a sensitivity analysis that modifies EPI’s family budget to more closely reflect the methodology of other family budgets. The findings were similar to those included in the body of this report.

²⁴ See **Appendix A** for an explanation of how “other necessities” was calculated.

²⁵ For an analysis of housing affordability challenges for older adults, see Samara Scheckler et al., “Older Adults Struggle to Meet the Dual Burden of Housing and Care,” Joint Center for Housing Studies, Harvard University, May 2025, <https://www.jchs.harvard.edu/research-areas/working-papers/older-adults-struggle-meet-dual-burden-housing-and-care>.

The Relationship Between Cost Burden and Shelter Poverty

Figure 1. How Shelter Poverty Rates Vary by Cost Burden



Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://web.archive.org/web/20240305013927/https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities. The shelter poverty rate for households who spend 0% of income on housing costs is high because households with zero housing costs can still be considered shelter poor if their income is less than their non-housing costs.

Understanding the relationship between shelter poverty and cost burden is useful for predicting when the two affordability measures are likely to intersect and when they are likely to diverge.

Figure 1 plots the shelter poverty rate (the number of shelter poor households/total households multiplied by 100) against cost burden. The shelter poverty rate generally increased as the percentage of income devoted toward housing increased. While households devoting 23%-29% of income toward housing costs are generally considered *not cost burdened*, **Figure 1** shows that the shelter poverty rates for these households exceed 25%. For households spending more than 30% of household income on housing—generally considered *cost burdened*—shelter poverty rates exceeded 50%.

Researchers have recognized that, in addition to the 30% standard, spending more than 50% of income on housing is a meaningful threshold for affordability. HUD, the National Low Income Housing Coalition, and the Joint Center for Housing Studies consider such households to be *severely cost burdened* and have examined the prevalence and characteristics of such households in their efforts to understand housing affordability in the United States.²⁶ Among severely cost burdened households, 100% experienced shelter poverty.

²⁶ HUD, *Worst Case Housing Needs: 2025 Report to Congress*, July 2025, <https://www.huduser.gov/portal/publications/Worst-Case-Housing-Needs-2025-Report-to-Congress.html>; National Low Income Housing Coalition, "The Gap: A Shortage of Affordable Homes," March 2025, pp. 10-19, <https://nlihc.org/gap>; and Peyton Whitney et al., "Housing Cost Burdens Climb to Record Levels (Again) in 2023," *Housing Perspectives* (blog), Joint Center for Housing Studies, December 2024, <https://www.jchs.harvard.edu/blog/housing-cost-burdens-climb-record-levels-again-2023>.

Figure 1 suggests that affordability measures will likely exhibit similar patterns for severely cost burdened households, as both measures indicate that these households are facing housing affordability challenges. In contrast, the measures may show different data patterns for households just below the threshold for being considered cost burdened. While the ratio approach assumes that housing costs are not burdensome for these households, the residual income approach suggests that a non-negligible portion are facing affordability challenges.

The Prevalence of Housing Affordability Challenges

This section examines how often the affordability measures agree and disagree about whether households are facing affordability challenges by estimating the prevalence of affordability challenges under both definitions. In 2023, there were fewer cost burdened households (housing costs exceed 30% of household income) than shelter poor households (housing and estimated non-housing costs exceed household income). There were 25.8 million (35.1%) cost burdened households (of whom 12.8 million were severely cost burdened households) and 32.0 million (43.6%) shelter poor households.

Table 2 reports the number and socioeconomic characteristics of non-elderly households that were cost burdened, shelter poor, neither, or both in 2023. Approximately 53.0% of all non-elderly households were neither housing cost burdened nor shelter poor, while 31.8% were both cost burdened and shelter poor.²⁷ While the two groups had similar median annual housing costs (\$18,000-\$18,700), households that were neither cost burdened nor shelter poor had four times as much median household income.

The housing affordability assessment for the remaining 15.2% of non-elderly households depended on the metric used. Approximately 11.9% of all non-elderly households were shelter poor but not housing cost burdened. That is, they did not have enough income to cover all of their estimated housing and non-housing costs but spent a relatively small share of income on housing. These households had more than twice as much residual income as households experiencing both housing cost burden and shelter poverty and were 1.6 times more likely to have children. The additional child care, food, and health care costs associated with having children may explain why these households were shelter poor despite having relatively low annual housing costs (a median of \$10,296).

Approximately 3.3% of non-elderly households were housing cost burdened but not shelter poor. These households devoted a large share of income toward housing costs but had enough income to cover their estimated housing and non-housing costs. They were high income (a median of \$92,000, compared to the \$30,500 median income of households that were shelter poor and cost burdened) and had a median residual income of approximately \$60,800. These households highlight one of the criticisms of the ratio approach: not accounting for differences in household income can lead to *overstating* housing affordability challenges for high-income households.

²⁷ Approximately 17.4% of households were severely cost burdened and shelter poor. No households were severely cost burdened but not shelter poor.

Table 2. Non-Elderly Households Experiencing Housing Cost Burden, Shelter Poverty, Neither, or Both in 2023

	Shelter Poor and Cost Burdened	Shelter Poor but Not Cost Burdened	Cost Burdened but Not Shelter Poor	Neither Shelter Poor nor Cost Burdened
Number of households	23.3 million	8.7 million	2.4 million	38.9 million
Percentage of households	31.8%	11.9%	3.3%	53.0%
Median household income	\$30,500	\$48,100	\$92,000	\$125,000
Median annual housing costs	\$18,024	\$10,296	\$30,924	\$18,696
Median residual income	\$13,760	\$37,680	\$60,760	\$105,340
Households with children	38.3%	60.0%	13.8%	33.5%
Households reported receiving SNAP benefits ^a	24.4%	22.0%	1.4%	2.8%
Households reported receiving health insurance through Medicaid or similar plans	41.5%	45.8%	4.9%	8.6%

Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities. A household was classified as having health insurance through Medicaid or a similar plan if it reported that someone in the household received health insurance through Medicaid, Medical Assistance, or any kind of government-assistance plan for those with low incomes or a disability.

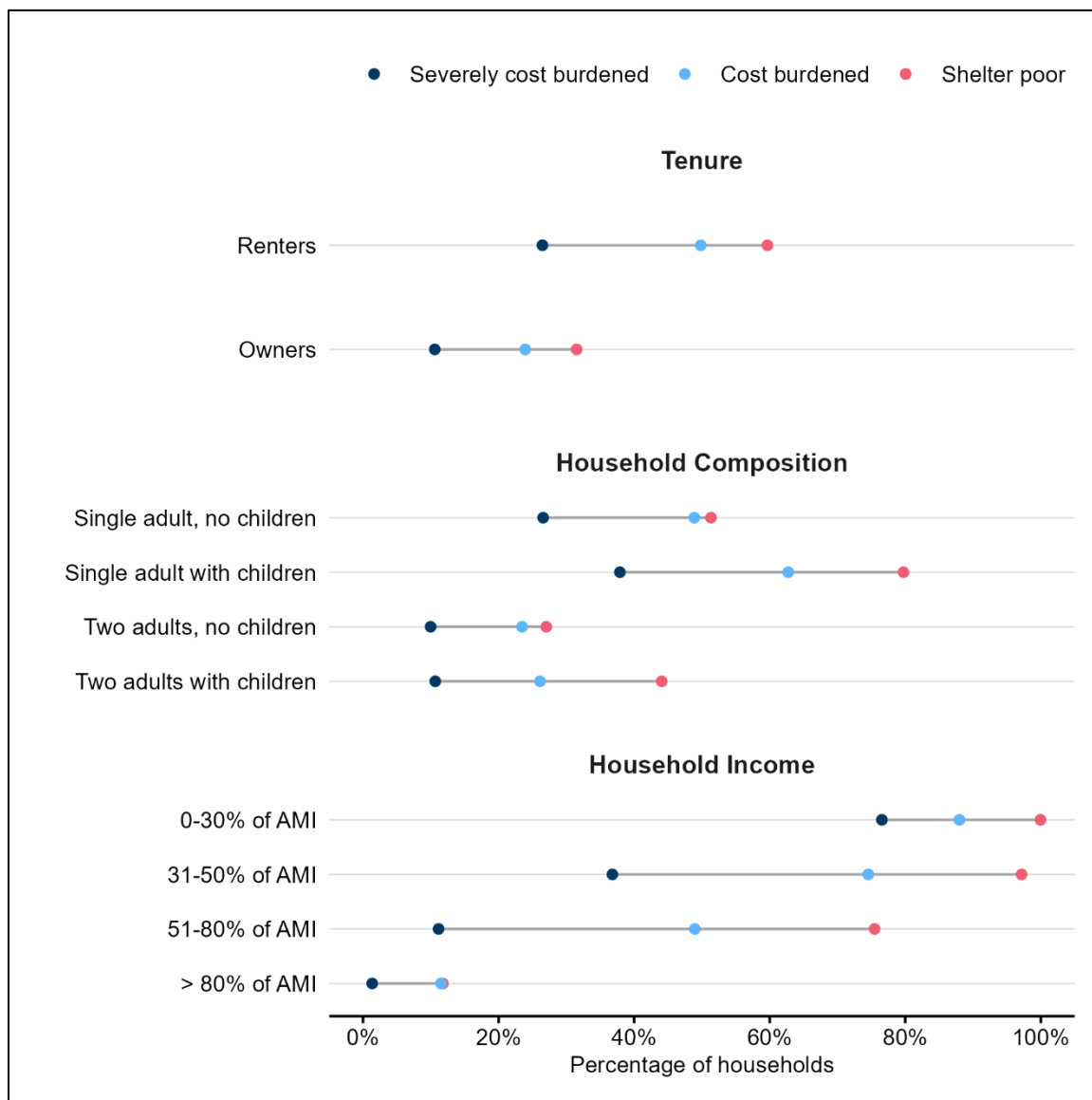
a. SNAP = Supplemental Nutrition Assistance Program.

Housing Affordability for Different Populations

The difference between what the ratio and residual income approaches consider “affordable” is illustrated when examining housing affordability patterns across different subpopulations. For many groups, the ratio approach estimates lower rates of housing affordability challenges than the residual income approach does. This section discusses how housing affordability challenges vary by affordability definition for non-elderly households with different sociodemographic characteristics. **Figure 2** and **Figure 3** show the percentage of severely cost burdened, cost burdened, and shelter poor households broken down by housing tenure, household income, and family type.

Figure 2. Percentage of Non-Elderly Households with Housing Affordability Challenges, by Household Characteristics

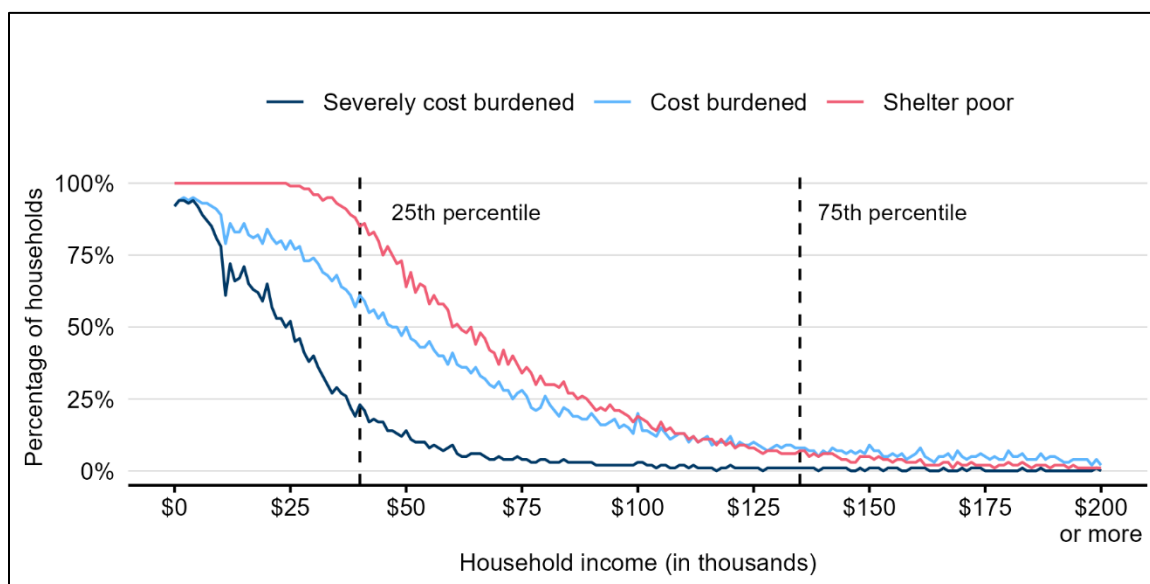
2023



Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities. AMI = Area Median Income.

Figure 3. Percentage of Non-Elderly Households with Housing Affordability Challenges, by Household Income
2023



Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.

Across all sociodemographic groups (except households in the top 25th percentile of the income distribution), the percentage of households facing shelter poverty exceeded the percentage facing cost burden. The magnitude of the gap between these percentages varied across groups:

- **Figure 2** shows that the gap between the percentage of cost burdened and shelter poor households was larger for renters (9.8 percentage points) than for homeowners (7.6 percentage points).
- **Figure 2** shows that the gap was larger for households with children (17.0-18.0 percentage points) than for those without children (2.5-3.6 percentage points).
- The gap also varied by household income (**Figure 2** and **Figure 3**).
 - While the gap for households above 80% of Area Median Income (AMI) was 0.3 percentage points, it was 12.0 percentage points for households at or below 30% of AMI, 22.6 percentage points for households at 31%-50% of AMI, and 26.5 percentage points for households at 51%-80% of AMI (**Figure 2**).²⁸

²⁸ Data on AMI were obtained from HUD and are based on the income limits used for the Section 8 program. The data are available at https://www.huduser.gov/portal/datasets/il.html#data_2023. Because Metropolitan Statistical Areas (MSAs) are not identified in the ACS PUMS, each household's Public Use Microdata Area (PUMA) was mapped to its MSA to determine its relative income category. The 1.8 million households whose PUMA spanned multiple MSAs (2.4% of the sample) were assigned to the lowest possible income category. For example, if a household's PUMA spanned two MSAs and the household was at 51%-80% of AMI in one MSA but above 80% of AMI in the other MSA, it was classified as being at 51%-80% of AMI. A sensitivity analysis where households were instead assigned to the (continued...)

- As shown in **Figure 3**, the gap between the percentage of cost burdened and shelter poor households ranged from 5 to 31 percentage points for households in the bottom 25th percentile of the income distribution (less than or equal to \$40,000) and from 0 to 4 percentage points for households in the top 25th percentile of the income distribution (greater than or equal to \$135,000).²⁹

The ratio and residual income approaches diverge the most for households with children and households with low income. This discrepancy arises because the ratio approach assumes that households need only 70% of income to purchase non-housing necessities regardless of household income or composition. The residual income approach considers that lower-income households and households that require child care may have insufficient income to purchase non-housing necessities even if they are devoting less than 30% of income toward housing.

Some additional takeaways about how housing affordability varies across sociodemographic groups were the following:

- For all affordability measures, the percentage of renters with affordability challenges was approximately double the percentage of homeowners with affordability challenges (**Figure 2**). Renters also made up the majority of households with housing affordability challenges according to all three affordability measures (**Table 3**).
- Among the household compositions examined, single adults with children had the highest percentage of households with affordability challenges (**Figure 2**). This was observed across all measures of affordability. While single adults with children made up 7.7% of all non-elderly households, they accounted for 13.8%-16.7% of those with housing affordability challenges (**Table 3**).
- The ratio approach suggests that having children is correlated with higher rates of housing cost burden for single-adult households but not for two-adult households (**Figure 2**). This may be because children create a need for additional bedrooms—and therefore additional housing costs—and, because two-adult households have more income than single-adult households do, the additional housing costs will likely represent a smaller increase in their housing-cost-to-income ratios. Thus, two-adult households are less likely than single-adult households to cross the 30% threshold when they have children.
- The lowest income families were the most likely to be shelter poor. While 11.8% of households above 80% of AMI were shelter poor, 75.5% of those at 51%-80% of AMI, 97.2% of those at 31%-50% of AMI, and 99.9% of those at 0%-30% of AMI were shelter poor (**Figure 2**).
- Compared to households in other relative income categories, households at 0%-30% of AMI faced the highest rate (76.5%) of severe cost burden (**Figure 2**) and made up the majority (64.0%) of households experiencing severe cost burden (**Table 3**).

highest possible income categories was conducted. The estimates were similar, and the qualitative findings were unchanged.

²⁹ **Appendix C** explores how cost burden and shelter poverty rates vary across states. It similarly finds that these rates diverge more for states with high percentages of residents in the bottom 25th percentile of the income distribution for non-elderly U.S. households.

Table 3. Distribution of Non-Elderly Households with Affordability Challenges

Household Characteristics	Percentage of Cost Burdened Households	Percentage of Severely Cost Burdened Households	Percentage of Shelter Poor Households	Percentage of All Households
Housing Tenure				
Renters	61.2%	65.5%	58.9%	43.1%
Owners	38.8%	34.5%	41.1%	56.9%
Household Composition				
Single adult with children	13.8%	16.7%	14.1%	7.7%
Single adult, no children	42.8%	46.9%	36.2%	30.7%
Two adults with children	22.2%	18.2%	30.1%	29.8%
Two adults, no children	21.2%	18.2%	19.7%	31.7%
Household Income				
0%-30% of AMI	36.5%	64.0%	33.4%	14.6%
31%-50% of AMI	20.9%	20.8%	21.9%	9.8%
51%-80% of AMI	23.2%	10.7%	28.8%	16.7%
> 80% of AMI	19.3%	4.6%	15.9%	58.9%

Source: Figure created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities. AMI = Area Median Income.

The Size of the Affordability Gap

One reason policymakers care about housing affordability measures is because such measures are used to determine federal housing subsidy levels. The next two sections detail (1) the amount of additional income households needed in 2023 for their housing to be considered “affordable” and (2) what an “affordable” housing cost would have been given households’ observed income. This information can be useful in determining which affordability measure to use when determining subsidy levels and how much assistance would be needed.

This section includes estimates of the *affordability gap*, which is the amount of additional income households would need to ensure that their housing was “affordable” under different affordability measures. For the ratio-based measures of affordability, the gap is the amount of additional income needed to ensure that the ratio of housing costs to household income did not exceed the specified percentage (30% or 50%). For the residual-income-based measure, the gap is the amount that would ensure that households had enough money to pay for housing and non-housing necessities.

Table 4 reports the amount of additional income households require to no longer be cost burdened or shelter poor given their current housing costs. It shows that the estimated additional income required to eliminate shelter poverty among non-elderly households was more than three times the amount required to eliminate cost burden and more than six times the amount required to eliminate severe cost burden. The average non-elderly household would have required an

additional \$9,665 in household income to no longer be considered housing cost burdened; \$10,257 to no longer be considered severely cost burdened; and \$27,448 to no longer be considered shelter poor.

The discrepancy between the amounts required for a household to no longer be considered cost burdened versus shelter poor is driven by mathematical differences between the ratio and residual income approaches. Consider a household with \$50,000 in income that pays \$20,000 for housing and requires \$50,000 to pay for non-housing goods according to the family budget. For housing to be “affordable,” the ratio approach requires this household’s income to increase by at least \$10,000 so that the ratio of housing costs to household income is at or below 0.3. In contrast, the residual income approach requires this household’s income to increase by at least \$20,000 so that household income is greater than or equal to the sum of housing and non-housing costs.

Table 4. Affordability Gap for Non-Elderly Households: 2023

	Number of Households (in Millions)	Total Affordability Gap (in Billions)	Mean Annual Gap
Cost burdened	25.8	\$248.9	\$9,665
Severely cost burdened	12.8	\$131.2	\$10,257
Shelter poor	32.0	\$879.2	\$27,448

Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute’s 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities. Very-low-income and extremely low-income households have household incomes at or below 50% and 30% of Area Median Income, respectively.

Housing Cost Levels “Affordable” to Households

This section considers what an affordable housing cost would be given a household’s current income. **Table 5** presents estimates of “affordable” annual housing costs under each affordability approach for households at different points of the income distribution. The first row in the table reports the 25th, 50th, and 75th percentiles of the income distribution for non-elderly U.S. households in 2023. The remaining rows display what would be considered the maximum affordable annual housing costs for each of these income levels under each affordability approach. For the ratio approach, maximum affordable housing costs are calculated as 30% of household income. For the residual income approach, maximum affordable housing costs are household income minus the mean non-housing costs for a household with the specified amount of household income.³⁰

³⁰ The mean non-housing costs for households at the 25th, 50th, and 75th percentile of the income distribution were \$41,519, \$58,211, and \$71,710, respectively.

Table 5. Annual Housing Costs Considered “Affordable,” by Affordability Approach and Income
2023

	Percentile of the Income Distribution		
	25 th	50 th	75 th
Household income	\$40,000	\$77,800	\$135,000
Approach to Measuring Housing Affordability	Maximum “Affordable” Annual Housing Cost		
Housing-cost-to-income ratio	\$12,000	\$23,340	\$40,500
Residual income	\$0	\$19,589	\$63,290
Approach to Measuring Housing Affordability	Income Left After Paying the Maximum “Affordable” Annual Housing Cost		
Housing-cost-to-income ratio	\$28,000	\$54,460	\$94,500
Residual income	\$40,000	\$58,211	\$71,710

Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute’s 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.

The ratio and residual income approaches diverge more when calculating affordable housing costs for households at the low and high ends of the income distribution compared to households at the middle. While there was a difference of less than \$4,000 in affordable housing costs for the median household, there was a difference of at least \$12,000 for households at the 25th and 75th percentile of the income distribution. The ratio approach estimates that households at the 25th percentile can pay *more*—and households at the 75th percentile can pay *less*—than what the residual income approach estimates that they can pay.

These differences reinforce one of the concerns about the ratio approach: It may understate affordability challenges for lower-income households and overstate challenges for high-income households. The lower-income household likely requires more than \$28,000 to pay its non-housing expenses, while the higher-income household probably requires less than \$94,500 to cover its non-housing expenses.

Another takeaway from **Table 5** is that the residual income approach allows for the possibility that a household with positive income may not be able to afford to pay *any* amount toward housing, while the ratio approach does not. Households at the 25th percentile of the income distribution cannot afford to pay anything toward housing under the residual income approach but can afford to pay up to \$12,000 under the ratio approach. By design, if a household has non-zero income, the ratio approach will always determine that it can pay some amount of income toward housing. Meanwhile, when non-housing costs are high and/or household income is low, the residual income measure can determine that any housing costs would be too high for a household in the absence of additional household income.³¹

³¹ In a separate analysis, CRS found that the residual income approach estimates that 99.9% of households at or below 30% of AMI could not afford to pay any amount of income toward housing.

Housing Insecurity

As discussed previously, one of the criticisms of the ratio and residual income approaches is that they ignore households' preferences for housing. Both measures set limits on how much households *should* spend on housing without accounting for the fact that some households may have housing-intensive preferences that make them *want* to spend more on housing costs than the measures indicate they should. One way to attempt to disentangle these two considerations is to look at measures of housing insecurity among families with estimated affordability challenges.

Using the 2023 AHS and the 2024 EPI FBC, CRS examined how the numerical measures of housing affordability intersected with households' self-reported experiences with housing insecurity. The AHS's 2023 Metropolitan Public Use File includes survey data on households in the 20 most populous metropolitan statistical areas (MSAs). From this file, CRS extracted housing tenure, household income, housing costs, and households' self-reported experiences with housing insecurity. The indicators of housing insecurity were the following:

- Difficulties affording housing payments in the past 12 months
- The number of missed housing payments in the past 12 months
- Being forced to move by a landlord, bank, government, or disaster in the past two years
- Moving to reduce housing costs within the past two years
- Being threatened with eviction in the past 12 months
- Receiving an eviction notice after being threatened with eviction in the past 12 months
- Receiving a foreclosure notice in the past 12 months
- A mortgage currently in foreclosure

To be compatible with EPI's FBC, this analysis also focuses on non-elderly households with one to two adults and zero to four children. Additional details on the methodology can be found in **Appendix A**.

As shown in **Table 6**, non-elderly households who were cost burdened, severely cost burdened, or shelter poor were more likely to report experiencing difficulty paying for housing than were households that were not cost burdened or shelter poor. **Table 6** reports the percentage of survey respondents who answered "Yes" to experiencing various indicators of housing insecurity. While 20% of households who were not cost burdened or shelter poor reported experiencing any difficulty affording housing costs in the past 12 months, 41%-42% of households experiencing some form of cost burden or shelter poverty had difficulty affording housing costs. Similarly, households experiencing some form of cost burden or shelter poverty reported finding it very difficult to afford housing and missing housing payments at rates more than four times higher than households that were not cost burdened or shelter poor. These rates did not vary much across housing affordability measures.

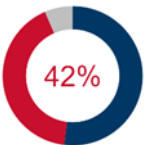
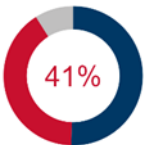
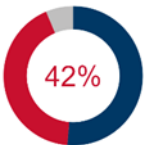
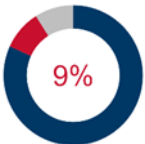
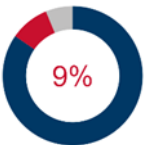
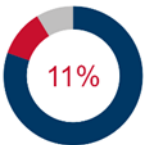
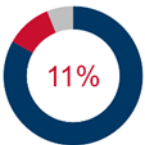
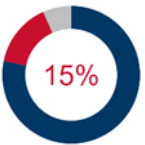
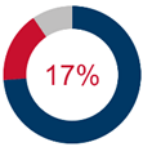
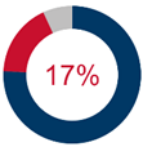
In contrast to the differences in reported difficulty paying for housing, there was little variation in the percentage of households that reported moving to reduce housing costs. The percentage of households that moved to reduce housing costs was 16% for those not considered cost burdened or shelter poor and ranged from 15% to 17% for households that experienced some form of cost burden or shelter poverty. Households that experienced some form of cost burden and/or shelter poverty may not have moved to reduce housing costs because there were few affordable housing

alternatives.³² Or these households may have chosen not to move because they have strong preferences for the amenities of their current housing.

The percentage of households that reported experiencing forced moves was higher among those that experienced some form of cost burden or shelter poverty. As shown in **Table 6**, 6%-7% of cost burdened and shelter poor households reported experiencing forced moves in the past two years compared to 3% of households that were not cost burdened or shelter poor. Similarly, the percentage of cost burdened and/or shelter poor households that were threatened with eviction or received foreclosure notices exceeded the percentage of not burdened and not shelter poor households experiencing these housing insecurity indicators.³³

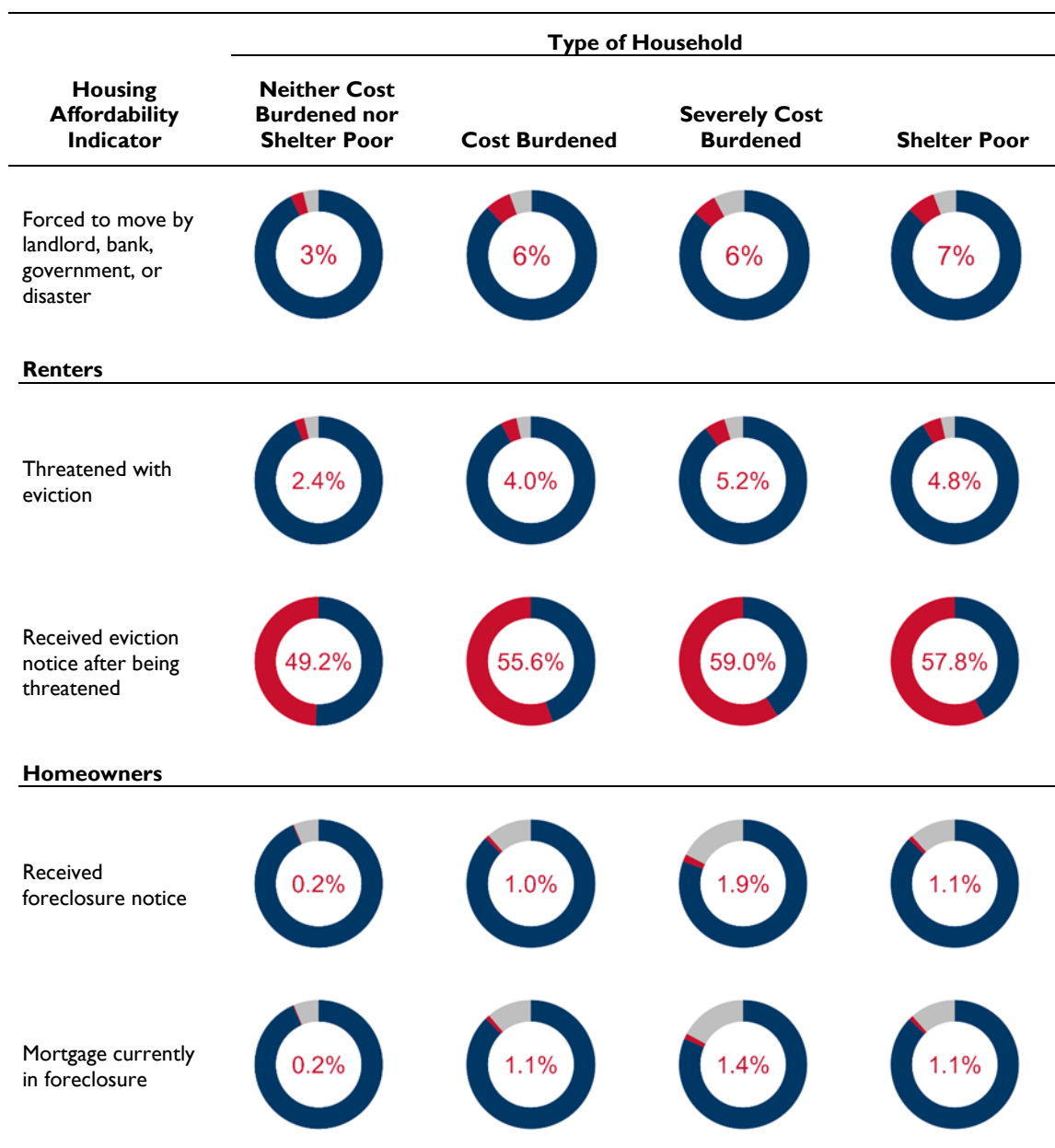
Table 6. Percentage of Non-Elderly Households Experiencing Indicators of Housing Insecurity

Table 7.2023

Housing Affordability Indicator	Type of Household			
	Neither Cost Burdened nor Shelter Poor	Cost Burdened	Severely Cost Burdened	Shelter Poor
All Households				
Any difficulty affording housing costs	 20%	 42%	 41%	 42%
Very difficult to afford housing costs	 1%	 7%	 9%	 7%
Missed housing payment	 2%	 9%	 11%	 11%
Moved to reduce housing costs	 16%	 15%	 17%	 17%

³² Researchers estimate that there is a housing supply shortage of approximately 4 million to 5 million units in the United States. Moreover, there is agreement that there is a shortage of affordable housing (according to the 30% housing-cost-to-income ratio rule) for the lowest-income renters. For more information, see CRS Report R48892, *Housing Supply: Current Trends and Policy Considerations*, by Lida R. Weinstock.

³³ Research has found that the AHS underestimates the prevalence of the evictions in the United States. Analysis of the 2017 AHS found that the survey estimated the national eviction rate at 0.8% in 2017 compared to the 2.3% eviction rate estimated by Princeton University's Eviction Lab. For more information, see Ashley Gromis and Matthew Desmond, "Estimating the Prevalence of Eviction in the United States," *Cityscape*, vol. 23, no. 2 (2021), pp. 279-290.



Source: Created by CRS using the American Housing Survey 2023 Metropolitan Public Use File, available at <https://www.census.gov/programs-surveys/ahs/data/2023/ahs-2023-public-use-file—puf-/2023-ahs-metropolitan-puf-microdata.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: Red = yes, blue = no, gray = did not respond. This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.

Households' Overall Economic Well-Being

Policymakers' interest in housing affordability is partially motivated by their interest in understanding how high housing costs affect households' overall economic well-being. The ratio approach assumes that housing costs are not impeding households' economic well-being if they do not exceed 30% of household income without consideration of a household's individual

circumstances. Conversely, the residual income approach attempts to identify and quantify households' overall economic well-being based on their individual circumstances. The following analysis uses the residual income approach to provide additional information on how housing costs affect households' ability to afford certain necessities.

According to the residual income approach, 43.6% of non-elderly U.S. households could not afford to meet all their basic needs with their household income. Focusing on the ability to pay for housing, transportation, health care, food, and child care with post-tax income,³⁴ CRS found that:

- 4.9% of households with children and 8.9% of households without children had no income left after paying for housing,
- 16.9%-23.7% of households with children would have had no income left after paying for housing and any two of the other four budget items (depending on which items are included),
- 18.1%-25.9% of households without children would have had no income left after paying for housing and any two of the other four budget items (depending on which items are included), and
- 43.2% of households with children and 31.1% of households without children would have had no income left after paying for all necessities in the household budget.

There were differences between renters and homeowners. CRS found that:

- 32.3%- 44.1% of renters with children would have had no income left after paying for housing and any two of the other four budget items compared to less than 13% of homeowners, and
- 69.1% of renters with children and 29.0% of homeowners with children would have had no income left after paying for all necessities in the household budget.

These findings are sensitive to the assumptions made about non-housing costs. There is no standard approach, and different methodologies can affect the findings. **Appendix B** presents results from a sensitivity analysis that illustrates how changing the assumptions made about what households are spending on non-housing necessities affects estimates of shelter poverty. The family budget was modified to (1) account for food and health care subsidies, (2) account for public transportation, and (3) change the amount households were assumed to be spending on "other necessities." After making these modifications to the family budget, the difference between cost burden and shelter poverty rates for various populations was smaller. Specifically, accounting for subsidies and public transportation reduced the difference between cost burden and shelter poverty rates for all non-elderly households, renters, owners, households with children, and households at every level of AMI.

Nonetheless, shelter poverty rates generally remained higher than cost burden rates when using the family-budget-based residual income approach. In contrast, a second sensitivity analysis showed that calculating non-housing costs as a percentage of the federal poverty line (rather than by using a family budget) can result in estimates of shelter poverty rates that are *lower* than the cost burden rate.

³⁴ Over 98% of households had post-tax income.

Policy Considerations

As demonstrated in this report, *how* housing affordability is defined can impact estimates of the scope of affordability problems and who faces housing affordability challenges. Moreover, each measure has its conceptual and practical advantages and disadvantages. The standard practice by lawmakers (and the general public) is to use the housing-cost-to-income ratio. Below are some policy considerations when deciding which housing affordability standard to use both in general and in specific contexts.

- **Purpose.** Lawmakers and researchers are often interested in housing affordability to understand households' experiences with housing instability. The ratio and residual income approaches produced similar estimates of the percentage of households experiencing various housing insecurity indicators. There can also be interest in how high housing costs affect a household's overall economic well-being. The residual income approach may be better suited for this analysis, because it attempts to directly estimate non-housing costs. For example, it is able to identify households whose housing costs inhibit their ability to cover the costs of their other basic needs regardless of the share of income devoted toward housing costs.
- **Feasibility.** One reason lawmakers are interested in housing affordability measures is because these measures are used to determine housing subsidy levels. A practical consideration for whether to use the residual income approach to measure housing affordability is whether it is *feasible* to alleviate shelter poverty. As shown in **Table 4**, it would have required \$879.2 billion in additional income for non-elderly U.S. households to no longer be considered shelter poor in 2023. Additional analysis found that 99.9% of households at or below 30% of AMI could not afford to pay any amount of income toward housing without becoming shelter poor. Their total affordability gap was \$312.1 billion. It may not be feasible to provide enough housing assistance to alleviate shelter poverty for non-elderly U.S. households even if assistance were restricted to the lowest income households.³⁵
- **Data availability.** Data availability imposes limitations on the residual income approach. The data needed to calculate the housing-cost-to-income ratio are available from reliably updated government sources, while the residual income approach requires external data that may not be regularly updated. Outdated data can lead to incorrect assessments of current affordability problems.
- **Inclusion.** Many of the existing family budgets do not include cost estimates for all household types (e.g., elderly and large households). Therefore, only a subsection of the U.S. population can be examined using the family-budget-based residual income approach absent further data collection and analysis, while the housing-cost-to-income ratio can be calculated for all household types using existing data.
- **Estimating costs.** As discussed previously, there can be wide variation in the cost of living proposed by different family budgets. This report utilizes EPI's FBC, which has been found to produce higher cost estimates than other family budgets

³⁵ The Congressional Budget Office estimated that the federal government spent \$50 billion on housing assistance for low-income households in 2014. See Natalie Tawil and Chad Chirico, *Federal Housing Assistance for Low-Income Households*, Congressional Budget Office, September 2015, <https://www.cbo.gov/publication/50782>.

do.³⁶ This reflects the subjectivity involved in determining what should be included in a family budget and how spending on each item should be calculated.

Use of the residual income approach by federal policymakers would likely require the creation of a standard federal family budget. Though the practice has been discontinued, throughout most of the 1900s the Bureau of Labor Statistics regularly published family budgets to aid policymakers in determining what would be considered a “modest, but adequate” budget for American families.³⁷

- **Multiple measures for research.** Policymakers and researchers may consider using multiple measures of affordability for research purposes. As discussed previously, it may not be financially feasible for policymakers to use the residual income measure when creating federal policy such as setting housing subsidy levels. However, the residual income approach can be incorporated into research to help policymakers better understand financial challenges facing U.S. households. This report finds that the ratio and residual income approaches produce dissimilar estimates for certain types of households, such as low-income households and households with children. Including the residual income approach as a supplementary housing affordability measure would allow for more thorough analysis of these populations.

³⁶ Joshi et al., *Measuring Economic Needs*, pp. 6-8.

³⁷ David S. Johnson et al., *A Century of Family Budgets in the United States*, Bureau of Labor Statistics, May 2001, pp. 29-38, <https://www.bls.gov/opub/mlr/2001/05/art3full.pdf>.

Appendix A. Methodology

This appendix explains the methodology used in this report for estimating housing cost burden and shelter poverty rates. The methodology closely resembles that of researchers from Harvard University's Joint Center for Housing Studies.³⁸

American Community Survey

The analysis primarily relies on data from the 2023 ACS 1-year Public Use Microdata Samples (PUMS). The following sample restrictions were applied to the data:

- Restricting the data to owner- and renter-occupied housing units, including those who are not paying rent (131,332,363 households)
- Excluding households where the head of the household is under age 18 (less than 1% of all households)
- Excluding households where the married or unmarried partner of the head of household is under age 18 (less than 1% of all households)
- Excluding households with members age 65 or older (19.6% of all renter households and 38.9% of all owner households)
- Restricting the data to households with one to two adults and zero to four children (86.5% of all renter households and 80.3% of all owner households)

These restrictions produced an ACS sample representing 73,390,393 households (55.9% of all U.S. households). Households with negative income were re-coded to have zero income. For renters, housing costs as a percentage of household income were calculated as gross rent divided by household income multiplied by 100. For owner-occupied units, housing costs as a percentage of income were calculated as selected monthly ownership costs divided by household income multiplied by 100. Households with zero income and positive housing costs were re-coded so that their gross rent (selected monthly ownership costs) as a percentage of household income was 101%.³⁹ Households were then classified as housing cost burdened if their gross rent as a percentage of household income was greater than 30%. Households with no housing costs were classified as not cost burdened.

Calculating shelter poverty rates involved merging the ACS with external data sources. Residual income was calculated by subtracting gross rent/selected monthly ownership costs from household income. Each household's non-housing costs were then estimated using data from EPI's 2024 FBC, Missouri Census Data Center's Geocorr 2022, U.S. Department of Agriculture's (USDA's) Thrifty Food Plan for June 2023, and National Bureau of Economic Research's TaxSim35. Cost estimates were obtained for six categories: transportation, health care, food, child care, taxes, and other necessities.

Transportation. Annual transportation costs were obtained from the FBC, which is in 2023 dollars. Transportation costs vary by family type and county. They were calculated using the cost of auto ownership, auto use, and transit use from the Center for Neighborhood Technology's Housing and Transportation Affordability Index as well as data from the Bureau of Labor Statistics' Consumer Expenditure Survey. The FBC assumes that all adults are working and commuting. The mileage traveled includes work trips for all adults in the household and social

³⁸ Whitney Airgood-Obrycki et al., "The Rent Eats First": Rental Housing Unaffordability in the United States," *Housing Policy Debate*, vol. 33, no. 6 (2022), pp. 1272-1292.

³⁹ The cap at 101% aligns with the Census Bureau's methodology, which caps gross rent as a percentage of household income at 101%.

trips for the first adult only (in two-adult households). EPI inflated the data to 2023 dollars using the regional transportation Consumer Price Index.⁴⁰

The FBC provides county-level cost estimates, but the ACS PUMS does not identify a household's county. (It identifies only a household's state.) Therefore, households' transportation costs (and other costs) were determined by calculating the expected value of their costs. Specifically, Geocorr 2022 was used to determine the share of each state's population that resided within each county according to the 2020 decennial census. The cost of transportation for a household was calculated as the population-weighted mean cost of transportation in its state of residence.⁴¹ Mathematically, this means the cost of transportation for household (*i*) in state (*s*) was calculated as an expected value using the following formula:

$$\mathbb{E}[\text{transportation}_i] = \sum_{c=1}^c p^c \times \text{transportation}_c$$

where $\mathbb{E}[\text{transportation}_i]$ is the expected cost of transportation for the household, p^c is the share of the population in the state (*s*) located in the county (*c*), and transportation_c is the cost of transportation in the county (*c*). This expected value approach was also used to determine health care, food, and child care costs for each household, because these costs were also available only at the county level.

Health care. Annual health care costs were obtained from the FBC and varied by family size and county. The FBC assumed that health insurance was purchased through the Affordable Care Act marketplace and includes two costs:

1. The monthly premium of the lowest-cost bronze plan in the rating area, assuming all adults are 40-year-old nonsmokers
2. The three-year average out-of-pocket costs from the restricted-use geocoded 2019, 2020, and 2021 editions of the Medical Expenditure Panel Survey

Food. Food costs were calculated using USDA's Thrifty Food Plan for June 2023 and adjusted for geographic price variation using the FBC. While other food plans are available, the thrifty plan was chosen to provide the most conservative cost estimates. The food plan provides national estimates of the monthly cost of food for individuals in various age brackets. Using the ages reported in the ACS, food costs were estimated for each individual household member. For people ages 12 and older, the cost was estimated as the average food cost for males and females in those persons' age brackets. After food costs were calculated for each household member, they were aggregated to obtain a household level estimate of food costs. The household-level monthly food cost was then adjusted for family size using USDA's recommended adjustment factors. Costs were then adjusted for county-level variation in food costs by multiplying the household-level monthly food cost by a county-level multiplier constructed by CRS. Using the FBC, the multiplier was equal to the food costs for a two-parent, two-child household in a county divided by the mean annual food costs for a two-parent, two-child household. Costs were then multiplied by 12 to estimate annual food costs.

Child care. Child care costs were imputed from the FBC's estimates of annual child care costs. These cost estimates were mostly obtained from the National Database of Childcare Prices (NDCP), which provides county-level data. Because data were not available for all counties, EPI supplemented the NDCP with data from Child Care Aware of America. EPI assumes that a family with one child needs care only for a four-year-old, a family with two children needs additional

⁴⁰ Bureau of Labor Statistics, *Consumer Price Index Program, CPI-All Urban Consumers (Current Series) Series ID CUUR0100SAT – CUUR0400SAT [database]*, <https://data.bls.gov/timeseries/CUUR0100SAT>.

⁴¹ Because these are average costs, they underestimate costs for people living in higher-cost areas of the state and overestimate costs for people living in lower-cost areas of the state.

care for an eight-year-old, and a family with three or more children needs care for a four-year-old, an eight-year-old, and a 12-year-old. EPI assumes that four-year-olds require full-time intensive care, eight-year-olds require afterschool and summer care, and 12-year-olds require summer care. Using these assumptions and EPI's reported child care costs for each family type and county, CRS estimated child care costs for children in three age brackets—age 5 and under, 6-11, and 12—in each county. CRS assumes children age 5 and under require full-time intensive care, children age 6-11 require afterschool and summer care, and children age 12 require summer care. Child care costs for a household were then calculated using the reported ages of the children in the ACS.

Taxes. Each household's total tax liability was calculated as the sum of its federal and state liabilities. These liabilities were calculated using TaxSim35 with the following methodology:

- Single-adult households are single filers, and two-adult households are joint filers.
- The household head is the primary wage earner/taxpayer.
- All household income is treated as wage and salary income and attributed to the primary taxpayer.
- The primary taxpayer's age is used for the "page" variable (age of primary taxpayer), and the other adult's age is used for the "sage" variable (age of spouse).
- The number of dependents is equal to the number of children (under age 18) in the household.
- The number of dependents under age 13 is equal to the number of children under age 13.
- Rent paid is the annual contract rent paid.
- The amount spent on child care is the expected cost of child care based on the previously described methodology.
- The remaining variables are zeros.

Other necessities. CRS had two approaches to calculating the cost of "other necessities." CRS first followed EPI's methodology and calculated spending on other necessities as 35.4% of food and housing costs. EPI used the 2022 Consumer Expenditure Survey to determine how much households in the 20th to 40th percentiles of the income distribution spent on items that are necessary for "a modest yet adequate standard of living."⁴² This calculation excluded spending on food, housing, child care, health care, taxes, and transportation, because those expense categories are calculated separately. Spending on other necessities was then divided by total spending on food and housing in the Consumer Expenditure Survey. The proportion was 35.4%. In a sensitivity analysis reported in **Appendix B**, CRS calculated spending on other necessities as 10% of spending on the other expenditure categories. This approach is consistent with the United Way of Northern New Jersey's ALICE Household Survival Budget and the University of Washington's Self-Sufficiency Standard.

The costs of transportation, health care, child care, food, taxes, and other necessities were summed to estimate the household's non-housing costs. Households were then classified as shelter poor if their total non-housing costs exceeded their residual income.

⁴² The items included in EPI's "other necessities" category are apparel, personal care, household supplies (including furnishings and equipment, household operations, housekeeping supplies, and telephone services), reading materials, and school supplies.

American Housing Survey

CRS's methodology for analyzing the 2023 American Housing Survey (AHS) was similar to the methodology for analyzing the ACS. The AHS was used to examine housing insecurity among cost burden and shelter poor households. Household income, housing costs, MSAs, households' experiences with housing insecurity, and other household characteristics were obtained from the AHS. The AHS calculates total monthly housing costs as the sum of the amount spent on mortgage payments, rent, utilities, property taxes, homeowner or renter insurance, homeowners or condominium association amounts, lot rent amounts, and routine maintenance costs.

To calculate the cost of non-housing necessities, CRS used EPI's FBC and the previously described methodology. There are two differences. First, county-level cost estimates from the FBC were aggregated up to the MSA level instead of the state level. Second, for households in MSAs spanning multiple states, their tax liabilities were the population-weighted expected tax liabilities across the states in their MSAs. For example, if a household lived in the Boston-Cambridge-Newton, MA-NH MSA, CRS calculated what its tax liability would be in Massachusetts and what it would be in New Hampshire. The household's estimated tax liability was then calculated as (share of the population in the MSA that lives in Massachusetts multiplied by the household's estimated liability in Massachusetts) + (share of the population in the MSA that lives in New Hampshire multiplied by the household's estimated liability in New Hampshire).

Appendix B. Sensitivity Analyses

This appendix reports the results of multiple sensitivity analyses conducted by CRS.

Changing Spending on Other Necessities

Because prior research has found that EPI's methodology usually produces a higher budget than other family budget calculators do, CRS conducted a sensitivity analysis where spending on "other necessities" was calculated as 10% of household spending on all other budget items (housing, food, transportation, health care, child care, and taxes) rather than the formula employed by EPI. This is the methodology used in the United Way of Northern New Jersey's ALICE Household Survival Budget and the University of Washington's Self-Sufficiency Standard. The results from this analysis can be found in the fifth and sixth columns of **Table B-1**.

Subsidies

EPI's FBC assumes that households are not receiving government subsidies and that all households have cars. For this sensitivity analysis, CRS relaxed these assumptions by accounting for households that received Supplemental Nutrition Assistance Program (SNAP) benefits, were insured by Medicaid, and do not own cars. This is meant to determine if the family budget is overstating how much income households require for non-housing necessities. Because researchers have found that the ACS underreports the number of households receiving SNAP and Medicaid subsidies, CRS conducted this analysis twice⁴³:

- First, CRS changed food and health care costs only for households that reported receiving these subsidies in the ACS.
- Second, CRS changed food and health care costs for households that reported receiving these subsidies and households that were likely eligible for these subsidies in 2023 based on their annual household incomes.

Because households that experience temporary losses of income can be eligible for SNAP and Medicaid, these estimates will likely still undercount the number of people who received these benefits.

SNAP

CRS accounted for SNAP receipt by subtracting the estimated annual amount of SNAP benefits a household received from its estimated food costs. The amount of SNAP benefits received was calculated as benefits per household in the household's state multiplied by 12.⁴⁴ This amount was

⁴³ See Bruce Meyer and Robert George, "Errors in Survey Reporting and Imputation and Their Effects on Estimates of Food Stamp Program Participation," Center for Economic Studies, April 2011, <https://www.census.gov/library/working-papers/2011/adrm/ces-wp-11-14.html>; and Robert Hest, "Tracking the Medicaid Undercount," State Health Access Data Assistance Center, January 2023, pp. 1-6, https://shadac-pdf-files.s3.us-east-2.amazonaws.com/s3fs-public/publications/Medicaid_Undercount_ACS_1.23.pdf.

⁴⁴ For robustness, estimates were also calculated while setting SNAP benefits equal to the maximum possible monthly benefit a household could receive, given its size and state, multiplied by 12. USDA reports that in FY2023, 36% of SNAP households actually received the maximum benefit. See Jordan W. Jones, "Supplemental Nutrition Assistance Program (SNAP)—Key Statistics and Research," USDA, updated August 31, 2026, <https://www.ers.usda.gov/topics/food-nutrition-assistance/supplemental-nutrition-assistance-program-snap/key-statistics-and-research>. The results were similar.

adjusted for family size using the ratio of maximum benefits for a given household size to maximum benefits for a two-person household.⁴⁵

CRS used USDA's income eligibility guidelines for gross monthly income to determine which households in the ACS were likely to be eligible for SNAP.⁴⁶ According to the ACS, 8.7 million non-elderly households received SNAP benefits in 2023. CRS estimated that a total of 14.5 million non-elderly households were eligible for SNAP benefits. This information was used to account for SNAP benefits in the second version of the analysis.

Medicaid

The ACS asks survey respondents whether each member of their households receive health insurance through Medicaid, Medical Assistance, or any kind of government-assistance plan for those with low incomes or disabilities. This variable was used to determine how many members in a household needed market-based insurance. CRS assumed that members who received Medicaid had zero health care costs. Thus, when a household had members who received Medicaid or were likely eligible for Medicaid, the household's health care costs were recalculated to include only costs for members receiving market-based insurance. CRS again used the health care cost estimates from EPI's FBC.

Income limits for Medicaid eligibility were obtained from KFF.⁴⁷ According to the ACS, 17.1 million non-elderly households had at least one member who received health insurance through Medicaid, Medical Assistance, or another kind of government-assistance plan for those with low incomes or disabilities in 2023. CRS estimated that 23.4 million non-elderly households were eligible for Medicaid in 2023. This information was used to account for Medicaid in the second version of the analysis.

Transportation

For households who reported not having vehicles, CRS recalculated their transportation costs as the estimated annual cost of a bus pass for all adults in the household. Using the American Public Transportation Association's (APTA's) 2022 Public Transportation Fare Database, CRS estimated the annual cost of a bus pass as the median monthly cost of a pass multiplied by 12.⁴⁸ Costs were then adjusted for geographic price differences using the ratio of transportation costs for a two-parent, two-child household in a state divided by the mean transportation cost for a two-parent, two-child household in EPI's FBC.

Results

Table B-1 reports the percentage of all non-elderly U.S. households that were cost burdened and shelter poor under different definitions of *shelter poverty*. When accounting for subsidies and public transportation, the shelter poverty rate became much closer to the housing cost burden rate.

⁴⁵ CRS used a two-person household as the denominator for this ratio, because USDA reports that in FY2023, the average SNAP household size was 1.9. See Mia Monokovic and Ben Ward, "Characteristics of SNAP Households: Fiscal Year 2023," USDA, 2025, <https://www.fns.usda.gov/research/snap/characteristics-fy23>.

⁴⁶ The guidelines are available at <https://fns-prod.azureedge.us/sites/default/files/resource-files/FY23-Income-Eligibility-Standards.pdf>.

⁴⁷ See KFF, "Medicaid/CHIP Upper Income Eligibility Limits for Children, 2000-2026, available at <https://www.kff.org/medicaid/state-indicator/medicaidchip-upper-income-eligibility-limits-for-children/>; Medicaid Income Eligibility Limits for Parents, 2002-2025, available at <https://www.kff.org/medicaid/state-indicator/medicaid-income-eligibility-limits-for-parents/>; and Medicaid Income Eligibility Limits for Other Adults, 2011-2025, available at <https://www.kff.org/medicaid/state-indicator/medicaid-income-eligibility-limits-for-other-adults/>.

⁴⁸ APTA's 2022 Public Transportation Fare Database is available at <https://www.apta.com/research-technical-resources/transit-statistics/fare-database/>.

Without subsidies, the estimated shelter poverty rate was 5.9-8.5 percentage points higher than the housing cost burden rate, depending on how *shelter poverty* was defined. With subsidies, the shelter poverty rate was 0.7-4.6 percentage points higher than the housing cost burden rate, depending on how *shelter poverty* was defined and what assumptions were made about subsidies. However, some of the key findings remained the same. These included the following:

- Renters were more likely than homeowners to face challenges with housing affordability.
- The percentage of households with children and low-income households (below 80% of AMI) that were cost burdened was lower than the percentage that were shelter poor.
- Over 90% of households at or below 30% of AMI were shelter poor, and 88.0% were housing cost burdened.

Table B-1. The Prevalence of Housing Affordability Challenges Among Non-Elderly U.S. Households After Accounting for Subsidies and Public Transportation

	Housing Affordability Measure				
	Cost Burdened	Shelter Poor		Shelter Poor (Spending on “Other Necessities” Is 10% of the Household’s Budget)	
		Without Subsidies	With Subsidies	Without Subsidies	With Subsidies
Subsidized population: households that reported receiving SNAP and/or Medicaid					
All Households	35.1%	43.6%	39.7%	41.0%	36.4%
Owners	23.9%	31.5%	29.4%	29.1%	26.8%
Renters	49.8%	59.7%	53.3%	56.6%	49.1%
Households with children	33.6%	51.4%	47.1%	49.2%	44.2%
Household income is > 80% of AMI	11.5%	11.8%	10.3%	9.0%	7.6%
Household income is 51%-80% of AMI	49.0%	75.5%	66.7%	69.5%	59.6%
Household income is 31%-50% of AMI	74.5%	97.2%	88.6%	96.7%	85.6%
Household income is 0%-30% of AMI	88.0%	99.9%	94.7%	100.0%	93.4%
Subsidized population: households that reported receiving or are likely eligible for SNAP and/or Medicaid					
All Households	35.1%	43.6%	39.2%	41.0%	35.8%
Owners	23.9%	31.5%	29.0%	29.1%	26.3%
Renters	49.8%	59.7%	52.7%	56.6%	48.5%
Households with children	33.6%	51.4%	46.3%	49.2%	43.2%
Household income is > 80% of AMI	11.5%	11.8%	10.1%	9.0%	7.3%
Household income is 51%-80% of AMI	49.0%	75.5%	66.1%	69.5%	58.7%
Household income is 31%-50% of AMI	74.5%	97.2%	87.8%	96.7%	84.7%
Household income is 0%-30% of AMI	88.0%	99.9%	93.5%	100.0%	91.9%

Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. The two definitions of *shelter poverty* differ based on the amount needed for “other necessities.” The first definition calculates this as 35.4% of spending on food and housing; the alternative definition calculates it as 10% of spending on all other household budget items.

Using the Poverty Threshold-Based Residual Income Approach

CRS conducted additional data analysis comparing the ratio approach to the residual income approach when non-housing costs are estimated as a percentage of the federal poverty threshold adjusted for state-level price variation.⁴⁹ Households were considered shelter poor if their residual incomes were less than a specified percentage of the federal poverty threshold. Because there is no theory supporting the use of a specific percentage, several cutoff points were used: 66%, 100%, 150%, and 200%.⁵⁰

CRS found that the percentage of the federal poverty threshold used to calculate non-housing costs impacted affordability estimates. The percentage of households facing shelter poverty were:

- 16.8% when non-housing costs are 66% of the federal poverty threshold,
- 21.9% when non-housing costs are 100% of the federal poverty threshold,
- 29.6% when non-housing costs are 150% of the federal poverty threshold, and
- 37.6% when non-housing costs are 200% of the federal poverty threshold.

The percentage of households facing shelter poverty and cost burden (35.1%) were equal when non-housing costs were 184% of the federal poverty threshold.

Notably, the poverty-threshold-based measure produces lower estimates of the percentage of households facing shelter poverty than the family-budget-based measure does. This may be due to the poverty threshold representing a minimum subsistence level, whereas family budgets aim to determine what households need to have a “modest yet adequate” standard of living.⁵¹ The average poverty threshold for a household was approximately \$22,000. In contrast, the family budget used in this report estimated that the average household needed approximately \$87,000 to pay for basic necessities. Therefore, the estimates of shelter poverty from the poverty-threshold-based approach should be seen as the minimum percentage of households facing housing affordability problems, while the family-budget-based approach represents the maximum percentage of households facing housing affordability challenges.

⁴⁹ The poverty threshold was adjusted for state-level price variation using the Bureau of Economic Analysis’s Regional Price Parities.

⁵⁰ Previous researchers have used two-thirds of the federal poverty threshold. See Nandinee K. Kutty, “A New Measure of Housing Affordability: Estimates and Analytical Results,” *Housing Policy Debate*, vol. 16, no. 1 (2005), p. 119.

⁵¹ EPI, “Family Budget Calculator,” <https://www.epi.org/resources/budget/>.

Appendix C. Geographic Differences in Affordability Estimates

Figure C-1 plots the percentage of households with housing affordability challenges by affordability measure and state (including the District of Columbia).⁵² In illustrating geographic differences in affordability estimates, **Figure C-1** shows that the percentage of shelter poor households was higher than the percentage of cost burdened households in all 50 states (but not in the District of Columbia). As with the variation across sociodemographic groups, the magnitude of the difference between the percentage of shelter poor and cost burdened households varied by state. There was:

- a difference of less than 1 percentage point in two states (California and Washington);
- a difference of less than 10 percentage points in 19 states;
- a difference of between 10 and 20 percentage points higher in 28 states; and
- a difference of more than 20 percentage points in three states (Alabama, Mississippi, and West Virginia).

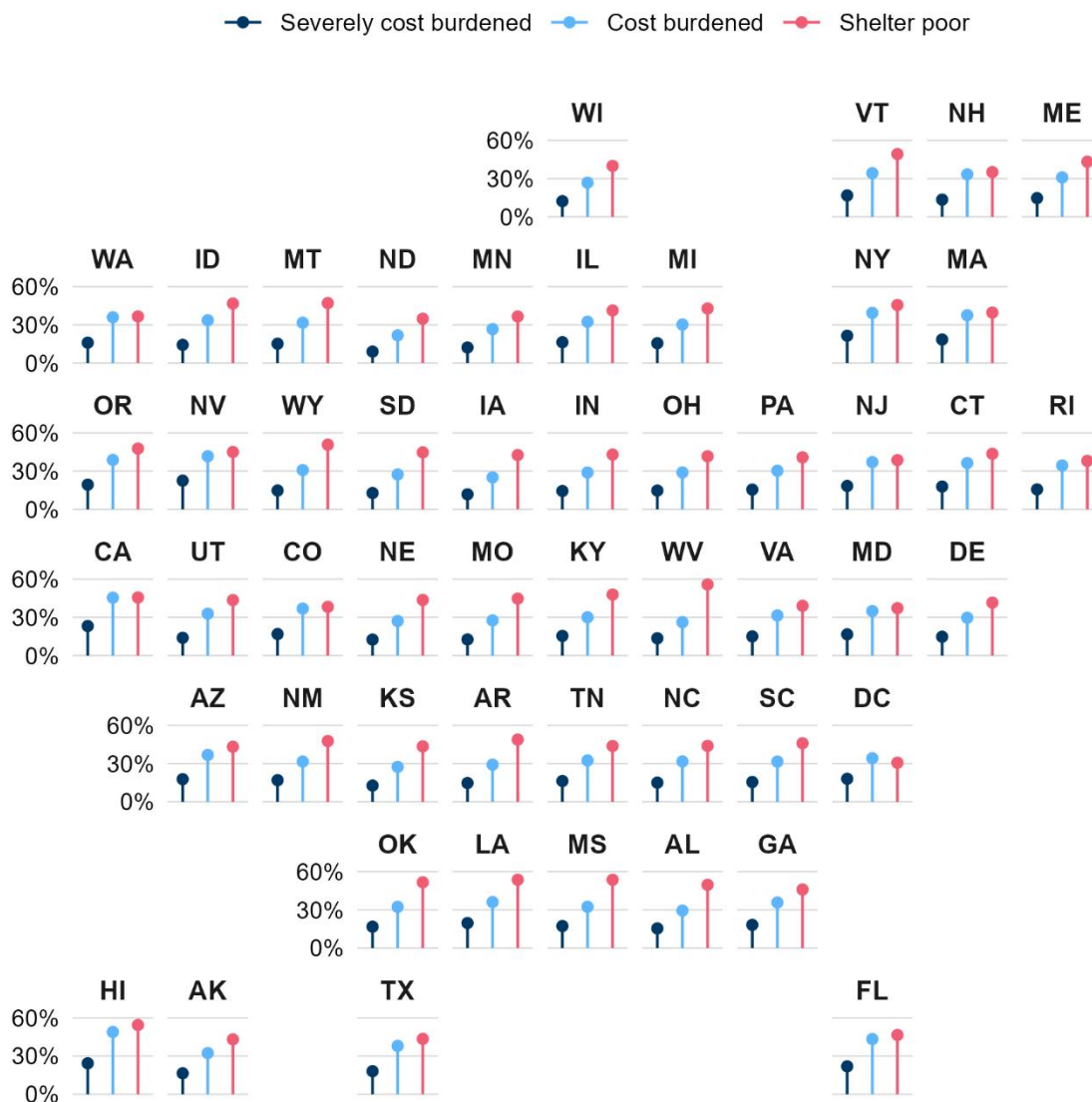
As shown in **Figure C-2**, the gap between a state's housing cost burden and shelter poverty rate is correlated with housing costs and household income. CRS calculated the gap between each state's housing cost burden and shelter poverty rate, then estimated the correlation between this gap and the state's mean monthly housing costs and percentage of households in the bottom 25th percentile of the U.S. income distribution. As shown in **Figure C-2**, as a state's mean housing cost increases, the gap between its shelter poverty and housing cost burden rate decreases. That is, states with relatively low housing costs have more differences between the percentage of cost burdened and the percentage of shelter poor households.

One explanation of this is that the ratio approach underestimates how much income households need to cover their non-housing costs. Even though their housing costs and subsequent housing-cost-to-income ratios are low, households' incomes may still be too low to cover all of their basic needs.

Figure C-2 also shows that states with higher percentages of households in the bottom 25th percentile of the U.S. income distribution also have larger gaps between their housing cost burdens and shelter poverty rates. This is consistent with CRS's earlier finding that rates of cost burden among low-income households—especially those in the bottom 25th percentile of the income distribution—were lower than rates of shelter poverty (**Figure 3**).

⁵² U.S. territories are not included in the ACS 2023 PUMS nor the FBC, so they were not included in this analysis.

Figure C-1. Percentage of Non-Elderly Households with Housing Affordability Challenges, by State
2023



Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.

Figure C-2. Characteristics Correlated with the Gap Between a State's Housing Cost Burden and Shelter Poverty Rates

2023



Source: Created by CRS using the 2023 American Community Survey 1-Year Public Use Microdata Sample, available at <https://www.census.gov/data/developers/data-sets/census-microdata-api.html>, and the Economic Policy Institute's 2024 Family Budget Calculator, available at <https://www.epi.org/resources/budget/>.

Notes: This analysis focuses on U.S. households with zero to four children and one to two adults who are all under age 65. Households are cost burdened if they devote more than 30% of household income toward housing costs and severely cost burdened if they devote more than 50%. A household is shelter poor if household income is insufficient to pay for housing and non-housing necessities. Non-housing costs include taxes, food, transportation, child care, health care, and other necessities.

Appendix D. Housing Affordability Measures

Measures of housing affordability seek to relate the prices of housing and non-housing goods to household financial resources in a meaningful way.⁵³ They can vary in terms of their mathematical approaches (a ratio versus a difference), whether they establish normative standards for housing and non-housing consumption, how they define *housing costs*, and how they define *household income*. In addition to the ratio and residual income approaches, common ways to conceptualize housing affordability include the following:

- **Family budget standard.** This approach involves specifying a basket of essential goods and services that should meet a household's basic needs. The acceptable quantity and quality of each item is usually determined using a combination of consumer expenditure surveys and expert opinions. Budgets are usually adjusted for both family size and geography.⁵⁴ As with the residual income approach, a shortcoming of this approach is that it relies on a societal standard for what households should spend on non-housing goods, ignoring differences in expenditure patterns that may be a result of preferences rather than financial strain. In addition, because the price of housing is highly variable, it is difficult to determine how much money households should have for housing. Even after specifying a geographic region and minimum physical standard, housing prices can exhibit high variance.⁵⁵
- **Subjective approach.** This approach considers housing affordable if a household chooses to pay for it given its budget constraints, meaning that no family lives in housing that is not affordable to it. It assumes that households are making optimal decisions given their preferences. This approach does not define a normative housing standard, addressing some of the conceptual issues with the residual income and family budget approaches. Under this approach the only households with housing affordability challenges are those experiencing homelessness, making it less useful than the other approaches when discussing topics such as housing insecurity. Another concern with this approach is that even though a household may choose to live in a unit, the unit may fail to meet standards that are generally considered socially acceptable or may inhibit the family's spending on other goods and services that are considered to have high social value. These considerations are reflected in opinion surveys that reflect high rates of concern about housing affordability⁵⁶ and therefore may be of importance to policymakers.

Numerous organizations have used the ratio, residual income, and family budget approaches to quantify the prevalence of housing affordability challenges in the United States. **Table D-1** provides an illustrative but inexhaustive list of housing affordability measures that have operationalized these concepts of housing affordability.⁵⁷

⁵³ This applies to the concept of affordability in general. For further discussion of different ways to think about affordability, see CRS In Focus IF13189, *Is "Affordability" An Attainable Policy Goal?*, by Lida R. Weinstock and Marc Labonte.

⁵⁴ A version of a family budget standard is used to establish benefit levels in SNAP.

⁵⁵ Stone, "What Is Housing Affordability?," pp. 151-184.

⁵⁶ See, for example, a survey by the Pew Research Center at https://www.pewresearch.org/politics/2025/04/23/economic-ratings-and-concerns-2025/pp_2025-4-23_trump-100-days_04-03/.

⁵⁷ Researchers at the Turner Center have also developed their own housing affordability measure that diverges from those discussed in this CRS report by relying on a more qualitative approach. For more information, see Daniel Shoag et al., "Affordability for Whom? Introducing an Inclusive Affordability Measure," Turner Center for Housing Innovation, May 2024, <https://turnercenter.berkeley.edu/wp-content/uploads/2024/05/Affordability-for-Whom-May-2024-.pdf>.

Table D-1. Housing Affordability Indicators

Indicator	Description	Reference
Ratio Approach		
30% housing-cost-to-income ratio	Housing is affordable if housing costs do not exceed 30% of household income. Housing costs are operationalized as gross rents (contract rent plus utilities) for renters and selected monthly owner costs for homeowners.	CRS Report R48450, <i>Housing Cost Burdens in 2023: In Brief</i> , by Maggie McCarty, Mark P. Keightley, and Mary Daniels, Peyton Whitney et al., “Housing Cost Burdens Climb to Record Levels (Again) in 2023,” <i>Housing Perspectives</i> (blog), Joint Center for Housing Studies, December 2024, https://www.jchs.harvard.edu/blog/housing-cost-burdens-climb-record-levels-again-2023 ; U.S. Census Bureau, “Nearly Half of Renter Households Are Cost-Burdened, Proportions Differ by Race,” press release, September 12, 2024, https://www.census.gov/newsroom/press-releases/2024/renter-households-cost-burdened-race.html .
Department of Housing and Urban Development (HUD) Gross Rent Affordability Index	The ratio of the median income for renter households in an area to the income needed to afford a rental unit priced at the median gross rent for the area while spending no more than 30% of income on rent.	Elaine Ng, “Housing Affordability Across the Country,” <i>PD&R Edge</i> , July 24, 2025.
HUD Homebuyer Affordability Index	The ratio of median income in an area to the income needed to purchase the median-priced home while spending no more than 30% of income on housing costs.	Ng, “Housing Affordability Across the Country.”
The Housing Wage	An estimate of the hourly wage a full-time worker must earn to afford a “modest” rental unit priced at HUD’s Fair Market Rent without spending more than 30% of income on rent.	Esther Colon-Bermudez et al., “Out of Reach,” National Low Income Housing Coalition, 2025, pp. 11-290, https://nlihc.org/oor .
New Renter Income Needed	The amount of annual household income needed to spend less than 30% of monthly income to newly lease a “typical rental” (a unit with rent equal to the monthly rental costs calculated by Zillow’s Observed Rent Index).	Anushna Prakash, “Methodology: Affordability Metrics,” Zillow, March 18, 2025, https://www.zillow.com/research/affordability-methodology-34975/ .

New Homeowner Income Needed	The amount of annual household income needed so that the total monthly payment on a newly purchased home with a 20% down payment does not exceed 30% of monthly income. The total monthly payment is calculated as a function of the mortgage, homeowner's insurance, property tax rate, and maintenance costs.	Prakash, "Methodology."
New Renter Affordability	The ratio of housing cost for a newly leased "typical rental" to household income for the median household where spending more than 30% of income on housing is considered unaffordable.	Prakash, "Methodology."
New Homeowner Affordability	The ratio of owner costs for a newly purchased home to household income for the median household where spending more than 30% of income on housing is considered unaffordable. Owner costs include the home's mortgage payment, homeowner's insurance, property taxes, and maintenance costs.	Prakash, "Methodology."
Location Affordability Measure	Housing is affordable if the joint cost of housing and transportation does not exceed 45% of income.	Center for Neighborhood Technology, "The Housing and Transportation Affordability Index," https://htaindex.cnt.org/ ; Kara Luckey, "Affordable for Whom? Introducing an Improved Measure for Assessing Impacts of Transportation Decisions on Housing Affordability for Households with Limited Means," <i>Research in Transportation Business and Management</i> , vol. 29 (December 2018), pp. 37-49.

Residual Income Approach

Residual income with a federal-poverty-line-based non-housing standard	Housing is affordable if the amount of income left after paying for housing (i.e., residual income) is greater than a specified percentage of the federal poverty line.	Matthew M. Brooks, "Measuring America's Affordability Problem: Comparing Alternative Measurements of Affordable Housing," <i>Housing Policy Debate</i> , vol. 33, no. 6 (2023), pp. 1293-1312; Nandinee K. Kutty, "A New Measure of Housing Affordability: Estimates and Analytical Results," <i>Housing Policy Debate</i> , vol. 16, no. 1 (2005), pp. 113-142.
--	---	--

Residual income with a family-budget-based non-housing standard	Housing is affordable if the amount of income left after paying for housing (i.e., residual income) is greater than or equal to the amount needed to purchase a minimally acceptable basket of non-housing goods. The amount needed for non-housing goods is determined using a family budget standard and varies with household size and geography.	Whitney Airgood-Obrycki et al., “‘The Rent Eats First’: Rental Housing Unaffordability in the United States,” <i>Housing Policy Debate</i> , vol. 33, no. 6 (2022), pp. 1272-1292; Christopher Herbert et al., “Measuring Housing Affordability: Assessing the 30 Percent of Income Standard,” Joint Center for Housing Studies, Harvard University, September 2018, https://www.jchs.harvard.edu/sites/default/files/media/imp/Harvard_JCHS_Herbert_Hermann_McCue_measuring_housing_affordability.pdf ; Luckey, “Affordable for Whom?”
---	--	---

Family Budget Standard

Family Budget Standard	Housing is affordable if household income is greater than or equal to the total cost of a household’s basic needs (e.g., housing, food, and transportation) at a minimally acceptable level adjusted for family size, composition, and geography. Housing costs are calculated using HUD’s Fair Market Rents, which estimate the 40 th percentile of gross rents for metropolitan statistical areas and non-metropolitan counties.	Economic Policy Institute, “Family Budget Calculator,” https://www.epi.org/resources/budget/ (accessed February 23, 2026); Massachusetts Institute of Technology, “Living Wage Calculator,” https://livingwage.mit.edu/ (accessed February 23, 2026); University of Washington, Center for Women’s Welfare, “Self-Sufficiency Standard,” https://selfsufficiencystandard.org/ (accessed February 23, 2026); United Way of Northern New Jersey, “Asset Limited, Income Constrained, Employed (ALICE) Household Survival Budget,” https://www.unitedforalice.org/ (accessed February 23, 2026).
------------------------	---	--

Source: Prepared by CRS.

Author Information

Mary Daniels
Innovation Policy Lab Fellow

Maggie McCarty
Specialist in Housing Policy

Mark P. Keightley
Specialist in Economics

Disclaimer

This document was prepared by the Congressional Research Service (CRS). CRS serves as nonpartisan shared staff to congressional committees and Members of Congress. It operates solely at the behest of and under the direction of Congress. Information in a CRS Report should not be relied upon for purposes other than public understanding of information that has been provided by CRS to Members of Congress in connection with CRS's institutional role. CRS Reports, as a work of the United States Government, are not subject to copyright protection in the United States. Any CRS Report may be reproduced and distributed in its entirety without permission from CRS. However, as a CRS Report may include copyrighted images or material from a third party, you may need to obtain the permission of the copyright holder if you wish to copy or otherwise use copyrighted material.